

Generalization: Properties of rewrite relations $\rightsquigarrow$

Assume a set A , and a binary relation $\rightsquigarrow \subseteq A \times A$. Then we define

For $a, a' \in A$ we say $a \rightsquigarrow^* a'$ if $a = a'$ or $\exists a'' \in A. [a \rightsquigarrow a'' \wedge a'' \rightsquigarrow^* a']$

$a \in A$ is in **normal form** if $\neg [\exists a' \in A. a \rightsquigarrow a']$

$\rightsquigarrow$ is **deterministic** if $\forall a, a_1, a_2 \in A. [a \rightsquigarrow a_1 \wedge a \rightsquigarrow a_2 \Rightarrow a_1 = a_2]$.

$\rightsquigarrow$ is **confluent** if $\forall a, a_1, a_2 \in A. [a \rightsquigarrow^* a_1 \wedge a \rightsquigarrow^* a_2 \Rightarrow \exists a' \in A. [a_1 \rightsquigarrow^* a' \wedge a_2 \rightsquigarrow^* a']]$

$\rightsquigarrow$ is **weakly normalizing** if $\forall a \in A. \exists d \in A. [a \rightsquigarrow^* d \wedge d \text{ is in normal form}]$

$\rightsquigarrow$ is **strongly normalizing** if $\forall a \in A. \text{there is no infinite sequence } a_1, \dots, a_n, \dots \text{ such that } \forall i \in \mathbb{N}. a_i \rightsquigarrow a_{i+1}.$

$\rightsquigarrow$ is **has unique normal forms** if $\forall a, a_1, a_2 \in A. [a \rightsquigarrow^* a_1 \wedge a \rightsquigarrow^* a_2 \wedge a_1, a_2 \text{ are both in normal form} \Rightarrow a_1 = a_2]$